DSAA 4011 · LAB 6 · FRIDAY 16 OCTOBER 2026

Teaching a Model to Reason

Way 2 of Lecture 6 inside a reasoning language model

A model is trained on questions and final answers only. Today: how it can still learn the **steps** between, in one tiny maze.

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Builds on Lecture 6: When the E-step is too big

Outline

- **The problem: the steps are hidden**

The model: a walker in a maze

The method: walk, keep, count

What it learns, and where it fails

The same loop in today's models

One question runs through the lab: how can a model learn steps it never sees?

Main Question

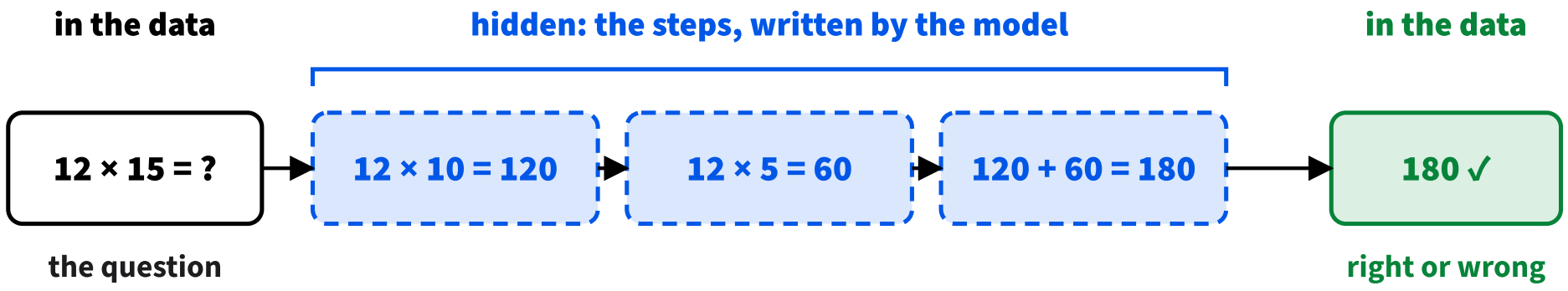
A reasoning model is trained on questions with answers only. Where does its reasoning come from?

1 · The problem: the steps are hidden

PROBLEM A model must learn good reasoning steps, but the data never shows them.

WE DO Take one example and mark what the data shows and what it hides.

WHY We must know exactly what is missing before we choose a method.



a real chain of thought has hundreds of steps, and each could have been written another way

SO **The steps are hidden, like the coins of Lecture 5, so they must be guessed.**

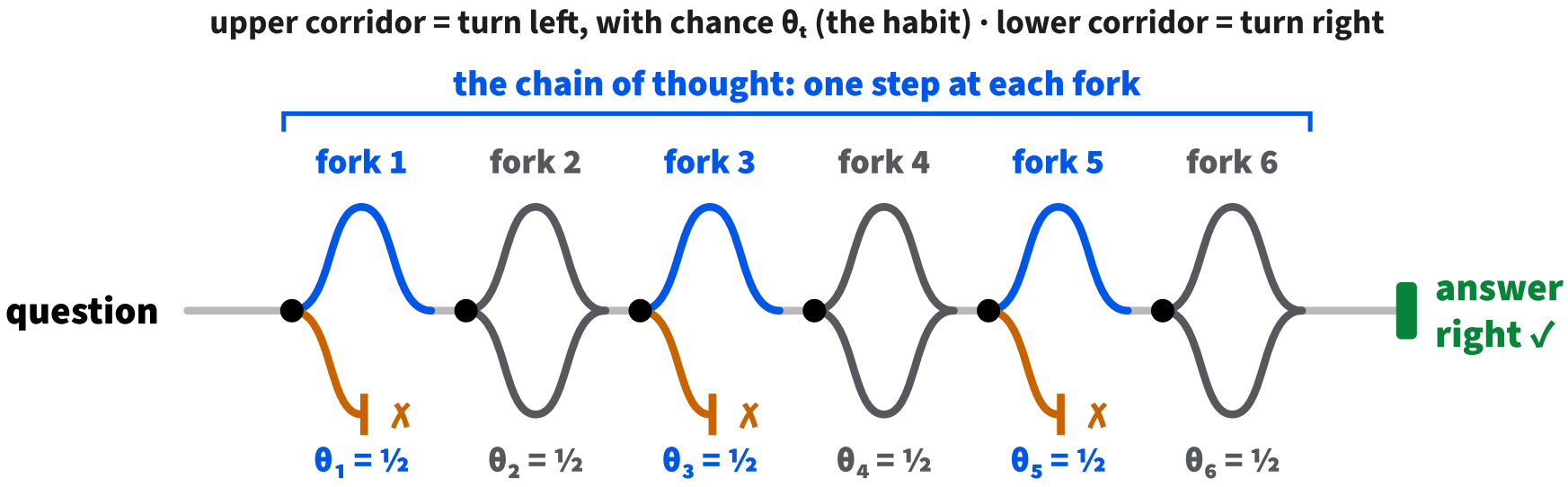
NEXT To watch the guessing closely, we first build a tiny model.

2 · The model: a walker in a maze

PROBLEM A real model is far too large to watch while it learns.

WE DO Build a walker in a maze of six forks, with one habit θ_t per fork.

WHY It keeps what matters: hidden steps, a right or wrong answer, weights to learn.



SO **Fork = reasoning step, exit = right answer, habits $\theta_1, \ldots, \theta_6$ = weights.**

NEXT To learn the habits, expectation–maximization (EM) weighs every path. How many paths are there?

3 · The method: what EM needs

PROBLEM The guess step of expectation–maximization (EM) needs the chance of every path, given the exit.

WE DO Count the paths: two choices at each of L steps give 2^L.

WHY The length of this list decides whether EM can run at all.

GUESS STEP for every path z, its chance given the exit, p(z | exit).

COUNT STEP count the turns at each fork, each path weighted by that chance.

steps L	paths 2^L	
6 (our maze)	64	a computer lists them at once
20	about a million	a short chain of thought
30	about a billion	a long chain of thought

SO **64 paths are easy; a real chain of thought has a billion, so the guess step cannot be written out.**

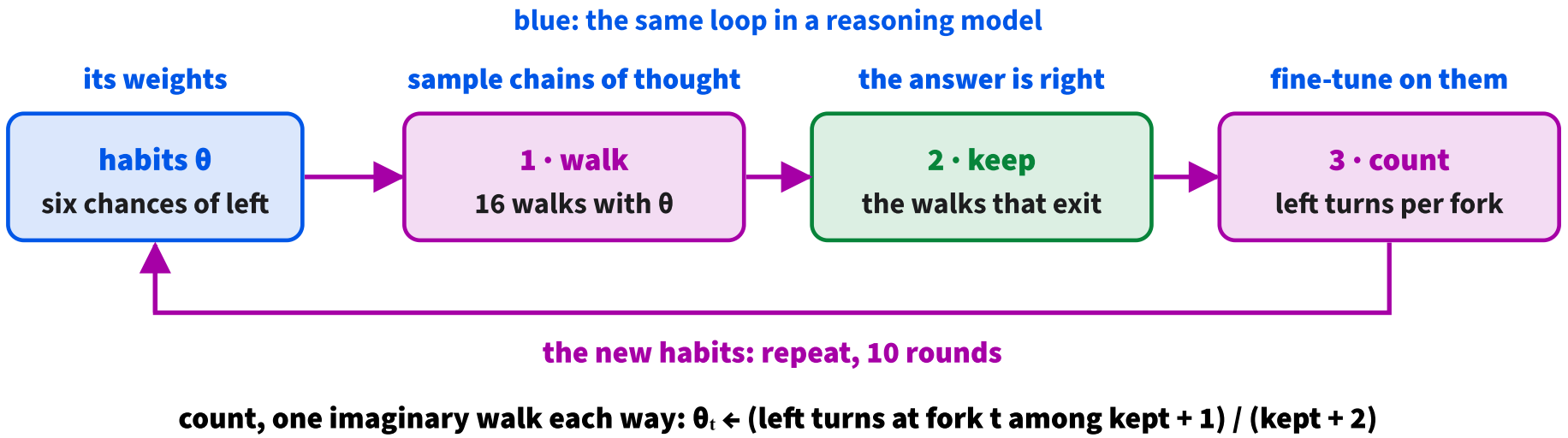
NEXT We need a way to learn that never writes the list.

3 · The method: walk, keep, count

PROBLEM The list of paths is too long to write out.

WE DO Walk many times, keep the walks that exit, and count their turns: Way 2.

WHY Running a model forward is the one thing every model does cheaply.



This is Monte Carlo expectation–maximization (EM) [Wei and Tanner, 1990].

SO **No list is needed: only walks and the exit check.**

NEXT But is a count of kept walks the same as EM's weighted count?

3 · The method: why the count is right

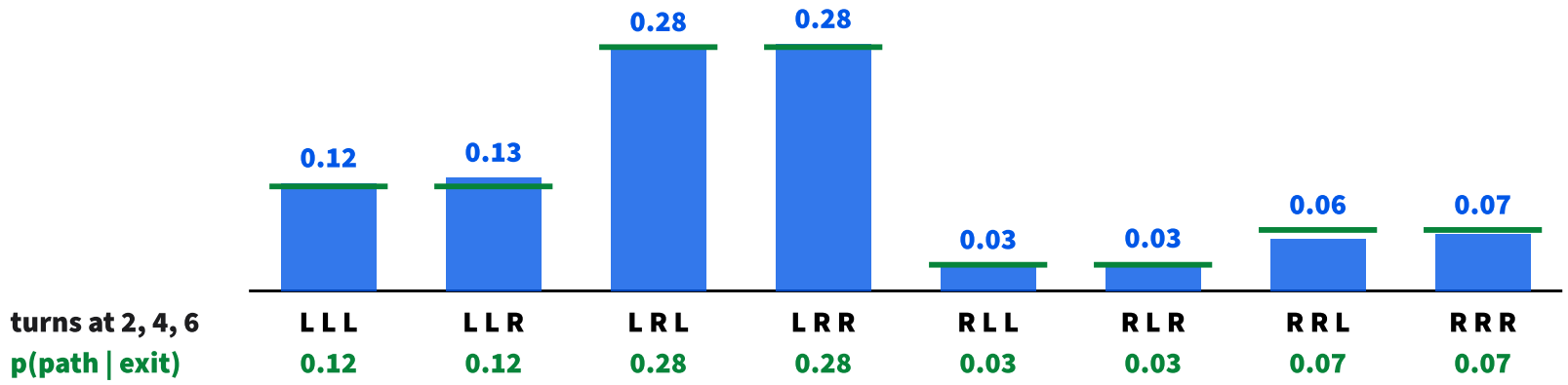
PROBLEM Walk, keep, count looks like a shortcut: is its count right?

WE DO Compute the chance that a kept walk is path z; test it on 16 000 walks.

WHY If kept walks follow p(z | exit), counting them is the count step of expectation–maximization (EM).

$$\Pr(\text{a kept walk is path } z) = \frac{p(z) \cdot \mathbf{1}[z \text{ exits}]}{\Pr(\text{exit})} = p(z \mid \text{exit})$$

habits: forks 1, 3, 5 at ½ · $\theta_2 = 0.8$, $\theta_4 = 0.3$, $\theta_6 = 0.5$ · 16 000 walks, 2 028 kept



blue bar: share among the kept walks · green line: p(path | exit), computed from the habits

SO **The kept walks follow p(z | exit): every bar meets its green line within 0.01 (Lecture 6, p. 29).**

NEXT The method is sound; now we run one round by hand.

4 · Round 1 by hand

PROBLEM The loop is still abstract.

WE DO Walk 16 times with every habit at ½, keep the walks that exit, count.

WHY One round by hand shows where every number comes from.

walk	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
fork 1	L	R	L	L	R	R	R	R	L	R	L	R	L	R	L	R
fork 2	R	L	R	R	R	R	L	L	R	L	L	L	L	R	L	L
fork 3	R	R	L	R	R	L	L	R	L	L	L	R	L	L	R	R
fork 4	R	R	L	R	L	L	L	L	L	R	L	L	L	R	L	L
fork 5	L	R	R	L	L	L	R	L	L	R	L	R	R	L	R	L
fork 6	L	R	L	R	R	L	L	R	L	L	L	R	L	L	R	R
exit?	✗	✗	✗	✗	✗	✗	✗	✗		✗		✗	✗	✗	✗	✗

kept: 2 of 16 walks exit (green) · fork 1: left in 2 of 2 → $\theta_1 = (2 + 1) / (2 + 2) = 0.75$

fork 2: left in 1 of 2 → $\theta_2 = (1 + 1) / (2 + 2) = 0.50$ · the same count at every fork

Q: How many of the 16 walks exit? A: About 2: a walk needs left at three forks, ½ · ½ · ½ = 1 in 8, and 16 / 8 = 2.

SO **Two walks are kept; every habit both of them agree on moves from ½ to 0.75.**

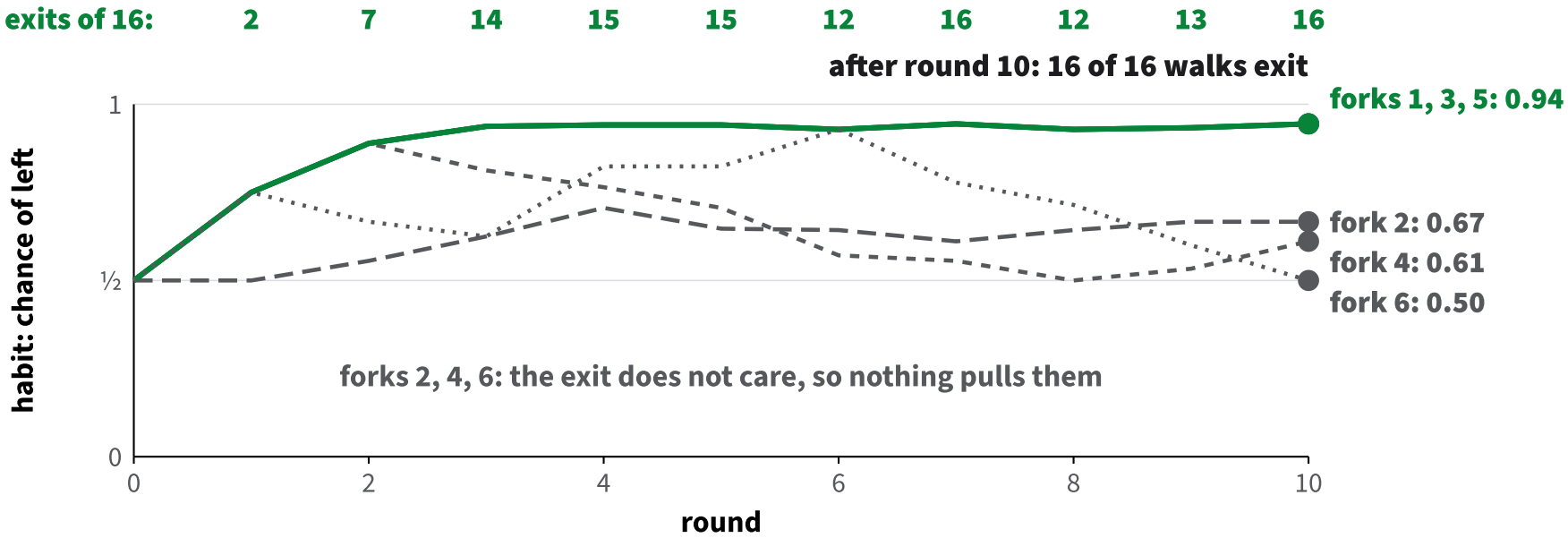
NEXT One round moves the habits a little. What do ten rounds do?

4 · Ten rounds

PROBLEM Two kept walks are too few to show what is learned.

WE DO Repeat the loop for ten rounds and follow all six habits.

WHY Learning shows in the trend over rounds, not in one round.



Q: Why do forks 2, 4, 6 not learn? A: Both turns reach the exit, so the answer check cannot tell them apart.

SO **Forks 1, 3, 5 rise to 0.94 and exits rise from 2 to 16 of 16; forks 2, 4, 6 only drift** **(p. 21)****.**

NEXT On this maze the loop works. When does it fail?

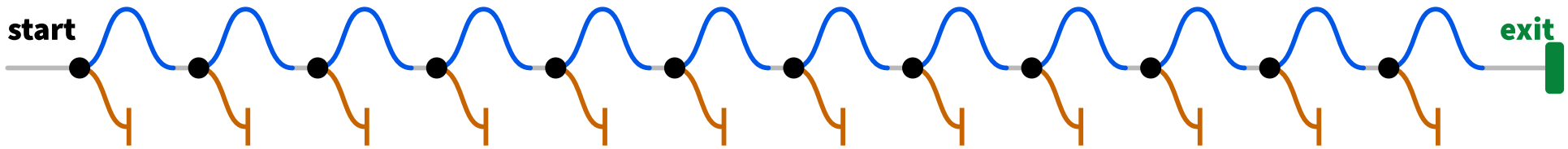
4 · Where it fails: a cold start

PROBLEM In a hard task a right answer needs many right steps.

WE DO Make all twelve forks matter and run the same loop.

WHY Real reasoning tasks often start where right answers are rare.

12 forks that all matter: the exit needs a left turn at every one



16 walks a round, 10 rounds: kept 0 0 0 0 0 0 0 0 0 0 of 16: nothing to count, the loop stalls

4 096 walks a round: kept 1, 31, 2823 in rounds 1, 2, 3: one lucky walk, then the loop moves

SO **One walk in 2^{12} = 4 096 exits, so 16 walks keep nothing and nothing is learned (Lecture 6, p. 57).**

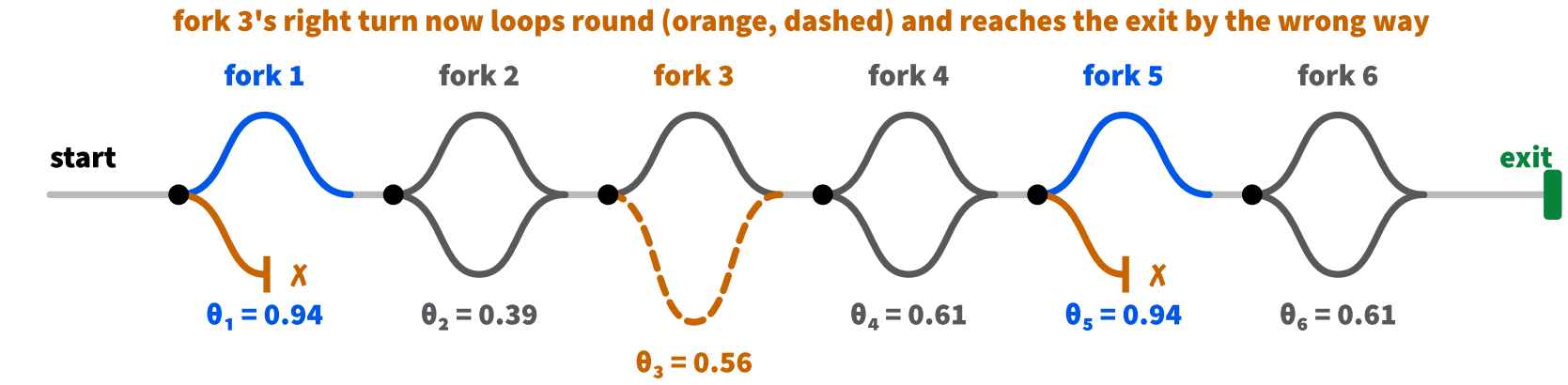
NEXT Here nothing is learned; next, something wrong is learned.

4 · Where it fails: a shortcut

PROBLEM The answer check rewards every walk that exits, however it got there.

WE DO Add a loop at fork 3 that also reaches the exit.

WHY Real models can also reach a right answer by wrong reasoning.



round 10: 16 of 16 walks kept, 7 of them through the loop; after 10 rounds $\theta_3 = 0.56$, not near 1

Q: How would you catch it? A: Check the steps too: keep a walk only if it turned left at fork 3; then $\theta_3 = 0.94$. This is process supervision [Lightman et al., 2023].

SO **Walks through the loop are kept, so fork 3 learns wrong reasoning, right answer.**

NEXT Both lessons carry over to the systems that train today's models.

5 · The same loop in today's models

PROBLEM Real reasoning models are far larger than the maze.

WE DO Match each system's training loop to walk, keep, count.

WHY The match tells us which lessons of the maze apply.

system	walk	keep	count
STaR: Self-Taught Reasoner	write one chain of thought per question	the chains whose answer is right	fine-tune on them; repeat
ReST-EM: Reinforced Self-Training	sample many chains per question	the chains that pass a check	fine-tune the base model; a few rounds
Llama 3	sample several answers per prompt	the best, by a learned judge	fine-tune; next round
DeepSeek-R1	sample chains from a trained model	the correct, readable ones	fine-tune, then more reinforcement learning

This is STaR [Zelikman et al., 2022] and ReST-EM [Singh et al., 2024]; Llama 3 [Grattafiori et al., 2024] and DeepSeek-R1 [DeepSeek-AI, 2025] run a rejection-sampling stage.

SO **All four run walk, keep, count, so both lessons apply: easy questions first and a check of the steps.**

NEXT We now gather the argument in three sentences.

Conclusion

TROUBLE The steps are **hidden**, and there are too many paths for the guess step of expectation–maximization (EM).

FIX Way 2: **walk, keep, count**. The kept walks follow p(path | exit), so counting them is the count step of EM.

RESULT The steps that decide the answer are learned; a cold start stalls; a shortcut is learned too, so **check the steps**.

Remember: a model learns only what the answer check can see.

Your turn

Predict, then run the maze.

Experiment 1: the maze walker (`python lab6_explore.py 1`, course site → Week 6 → Lab files)

- 64 walks a round instead of 16: do the free forks wander less?
- Fork 1 starts at habit 0.1: how many rounds until it is found?

Other problems, experiments 2 to 6

2. A noisy picture: Ways 2, 3 and 4.
3. The bits in your phone: messages.
4. Reading a library: guess alone.
5. Ranking players: one match at a time.
6. Messages that swing: damping.

Tip: write your guess first; surprises teach.

Selected References

DeepSeek-AI (2025).
DeepSeek-R1: incentivizing reasoning capability in large language models via reinforcement learning.
arXiv:2501.12948.

Grattafiori, A., Dubey, A., Jauhri, A., et al. (2024).
The Llama 3 herd of models.
arXiv:2407.21783.

Lightman, H., Kosaraju, V., Burda, Y., et al. (2023).
Let's verify step by step.
arXiv:2305.20050.

Singh, A., Co-Reyes, J. D., Agarwal, R., et al. (2024).
Beyond human data: scaling self-training for problem-solving with language models.
Transactions on Machine Learning Research.

Wei, G. C. G. and Tanner, M. A. (1990).
A Monte Carlo implementation of the EM algorithm and the poor man's data augmentation algorithms.
Journal of the American Statistical Association, 85(411):699–704.

Zelikman, E., Wu, Y., Mu, J., and Goodman, N. D. (2022).
STaR: bootstrapping reasoning with reasoning.
In Advances in Neural Information Processing Systems 35.

Thank you!

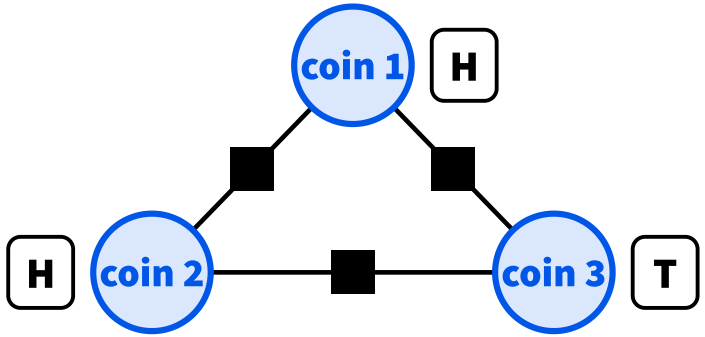
Next: Lecture 7. Way 2 returns whenever a model can only be run forward

Backup: the ring, which way can run, the count by hand, the imaginary walks, exact against sampled, other problems.

Backup: Lecture 6's ring

Eight stories were listed; Way 2 is the one that needs no list.

hidden: three coins, A or B
seen: the flips H, H, T
links: keep ¾ on every link



the question: coin 1's chance of being A



Look: Way 2 replayed the game, kept the runs that showed H, H, T and counted **(Lecture 6, p. 29)**; it is noisy but right on average.

Backup: which way can run?

Only Way 2 runs when the only tool is the model.

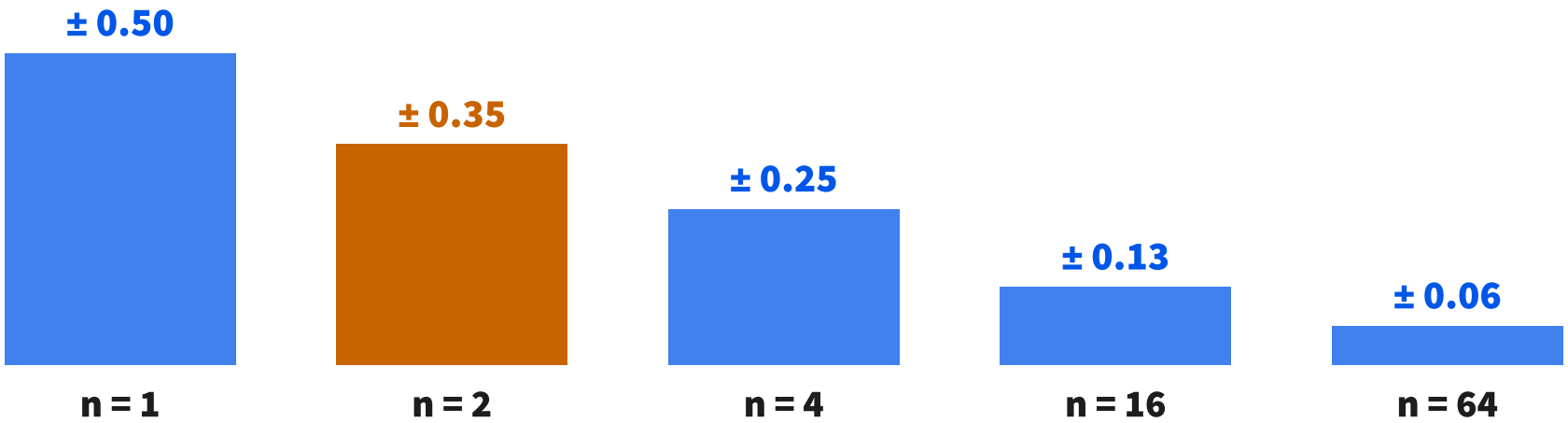
way	what it needs	in a reasoning model?
1 · messages along a line	a line: each step touches only the step before	**no: there is no line**
2 · act it out	running the model forward, and the answer check	**yes: run it forward**
3 · everyone guesses alone	each step touching only a few others	**no: each touches all**
4 · messages in a circle	small links between a few steps	**no: each touches all**

Read: a language model writes the next step from everything written so far, so every step touches all earlier ones: no line for Way 1 **(Lecture 6, p. 27)**, no small links for Ways 3 and 4.

Backup: why the imaginary walks

Two kept walks make a noisy count: add one imaginary walk each way.

typical error of a counted chance (truth ½) from n kept walks: √(½ · ½ / n)



round 1, fork 6: both kept walks turned left: raw count 2 / 2 = 1.00, as if fork 6 were sure

with one imaginary walk each way: $\theta_6 = (2 + 1) / (2 + 2) = 0.75$, held back from certainty

Read: by Theorem 7 **(Lecture 6, p. 31)** the error falls like $1/\sqrt{n}$, so two walks say little; and a habit of exactly 1 could never change again.

Backup: exact EM against sampling

Exact expectation–maximization (EM) over all 64 paths agrees where it matters.

	exact EM, all 64 paths		walk, keep, count	
fork	**round 1**	**round 10**	**round 1**	**round 10**
1, 3, 5 (matter)	0.75	**0.93**	0.75	**0.94**
2 (free)	0.50	0.50	0.50	0.67
4 (free)	0.50	0.50	0.75	0.61
6 (free)	0.50	0.50	0.75	0.50

Read: exact EM uses the same count, with the kept walks replaced by their average over all 64 paths: the forks that matter agree, and the free forks stay at ½ while sampling wanders, the noise of Theorem 7 **(Lecture 6, p. 31)**.

Backup: the count step of round 1 by hand

Round 1: two kept walks, L R L L L L and L L L L L L.

$$\theta_t \leftarrow \frac{\text{left turns at fork } t \text{ among the kept walks} + 1}{\text{kept walks} + 2}$$

> the count step (Lecture 6, p. 30)

$$\theta_1 = \frac{2+1}{2+2} = 0.75, \qquad \theta_2 = \frac{1+1}{2+2} = 0.50$$

> fork 1: both turned left; fork 2: one did

$$\theta_3 = \theta_4 = \theta_5 = \theta_6 = \frac{2+1}{2+2} = 0.75$$

> both turned left at forks 3 to 6

$$\text{without the imaginary walks: } \theta_1 = \frac{2}{2} = 1.00$$

> a habit at 1 could never move again

Notice: the exact guess step gives 1 at forks 1, 3, 5 and ½ at 2, 4, 6; two walks cannot tell fork 3 from fork 4.

Backup: Theorem 7's error for n kept walks

A counted chance p from n kept walks is off by about √(p(1 − p)/n).

kept walks n	p = ½: typical error	p = 0.9: typical error	where it happens in the deck
1	± 0.50	± 0.30	the cold start with 4 096 walks, round 1
2	± 0.35	± 0.21	round 1 (p. 9)
4	± 0.25	± 0.15	the shortcut maze, round 1 (p. 12)
16	± 0.13	± 0.08	rounds 7 and 10 (p. 10)
64	± 0.06	± 0.04	the kit, 64 walks a round

Notice: four times the kept walks halve the error (Lecture 6, p. 31); the free forks wander by about this much every round, which is the wandering seen over ten rounds.

Backup: five other troubles, in the kit

Five other troubles, in the kit as experiments 2 to 6.

	Trouble	Fix	Where
2	a noisy picture	redraw, guess alone or messages (Ways 2–4)	photos, scans
3	the bits in your phone	messages in a circle (Way 4)	phones, Wi-Fi, space
4	reading a million documents	guess alone, then count (Way 3, then count)	search
5	ranking players from wins	the truth judges one match at a time	matchmaking
6	messages that swing	damping: half old, half new	any cycles

Read: each is one of Lecture 6's four ways on a deployed system; the kit's README states a prediction for each, and the answers after it.